

Einstein relation and thermodynamic uncertainty relation

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1 Einstein relation

We consider a colloid with position $\mathbf{X}(t)$ and submitted to a force $\mathbf{F}(t)$. The position obeys a stochastic dynamics, which we do not know precisely, but which we characterize using two quantities. The first is the mobility $\boldsymbol{\mu}$, which is defined by the long time velocity acquired by the colloid under the application of a constant force $\mathbf{F}$, starting at time 0:

$$\langle \mathbf{X}(t) \rangle \underset{t \rightarrow \infty}{\sim} \boldsymbol{\mu} \cdot \mathbf{F}t. \quad (1)$$

The mobility is a tensor, whose component $\mu_{\alpha\beta}$ indicate the response in the direction α to a force in the direction β . The second is the diffusion tensor $\mathbf{D}$, which is defined in the absence of forces by

$$\langle \Delta X_\alpha(t) \Delta X_\beta(t) \rangle = \langle [X_\alpha(t) - X_\alpha(0)][X_\beta(t) - X_\beta(0)] \rangle \underset{t \rightarrow \infty}{\sim} 2D_{\alpha\beta}t. \quad (2)$$

At equilibrium, Einstein showed that the mobility and the diffusion coefficient are related by

$$\mathbf{D} = T\boldsymbol{\mu}. \quad (3)$$

Our goal is to prove this relation using the fluctuation-dissipation theorem.

We define the correlation as

$$C_{\alpha\beta}(t) = \langle X_\alpha(t) X_\beta(0) \rangle. \quad (4)$$

1. Express the correlation between the displacements (Eq. (2)) as a function of the correlation $C_{\alpha\beta}(t)$. Why does it require the presence of a trap to be well defined?
2. Express the diffusion coefficient with the time derivative of the correlation. You can assume that the long time limit and the time derivative commute.
3. What is the force conjugated to the observable X_α ?
4. Write the average position at time t with the time integral of a response function $R_{\alpha\beta}(t)$.
5. Relate the mobility to the response function introduced above.
6. Use the fluctuation-dissipation theorem to prove the Einstein relation.

2 Cost of precision of a biased random walk

We consider a continuous time random walk on the integers with biased jump rates k_+ to the right and k_- to the left. We define the affinity as $A = \ln(k_+/k_-)$. We consider a particle starting from $x = 0$ at $t = 0$.

2.1 Relative uncertainty of the final position

7. Write the position of the particle $x(t)$ as a function of the number of jumps $n_\nu(t)$ in the direction $\nu \in \{-, +\}$.
8. What is the distribution of the number of jumps $n_\nu(t)$? What are its mean and variance?

9. Deduce the mean position and the variance of the position and give the relative uncertainty of the final position,

$$\epsilon^2 = \frac{\langle [x(t) - \langle x(t) \rangle]^2 \rangle}{\langle x(t) \rangle^2}. \quad (5)$$

2.2 Entropy production

10. Write the average total entropy production rate during the process, $\langle \dot{S}_{\text{tot}} \rangle$ as explicitly as possible.
11. Separate a contribution that depends only on the rates k_ν and a contribution that would be zero if the probability density was uniform (which is possible on a circle).
12. Using the inequality $\ln(u) \leq u - 1$, show that the later is positive, providing a lower bound for the total entropy production rate.
13. The total entropy production at time t is $\Delta S_{\text{tot}} = t \dot{S}_{\text{tot}}$. Use the results above to obtain a lower bound for $\epsilon^2 \Delta S_{\text{tot}}$, which depends only on the affinity A , and a finite lower bound that does not depend on A . We give $x \coth(x) \geq 1$. When is this bound saturated?